Hein Zarni Aung

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EDUCATION

Stony Brook University Expected Graduation: May 2027

B.S., Computer Science + Applied Math & Statistics (Data Science) | GPA: 3.40

Honors & Scholarships: Global Excellence Scholarship

Relevant Coursework: Algorithms, Systems Fundamentals I & II, Statistics & Probability

RESEARCH

Vertically Integrated Project — Automotive Ethics Lab, Stony Brook University Starting Fall 2026

Team B: Simulations (5 people) | Faculty Lead: Amin Shirangi | Multi-semester, through graduation

- Joining the simulations sub-team of a Vertically Integrated Project studying automotive ethics, building and analyzing driving/traffic scenario simulations to study how autonomous vehicles should make decisions in the moment.

Undergraduate Research Assistant — Ethos Lab, Dept. of Computer Science, Stony Brook University Sep 2025 – Dec 2025

PI: Abisheka Pitumpe | Team: 2

- Built a pipeline to monitor newly issued SSL/TLS certificates in real time and flag potentially fraudulent domains impersonating EZ-Pass tolling services, scraping keywords from the raw HTML, CSS, JS, and URLs of newly created sites.
- Used Selenium to automate certificate and domain scraping and Python to filter and triage flagged sites across hundreds of candidates, auto-capturing homepage screenshots to speed up root-cause review and separate real fraud from noise.

INTERNSHIPS

Software Engineering Intern — Prenora Dynamics LLC Jun 2026 – Sept 2026 (expected)

~8-person team

- Own end-to-end design decisions for an in-house, in-platform video chat feature, tasked with researching and evaluating available SDKs/approaches to replace reliance on third-party platforms (e.g., Zoom, Google Meet).
- Set up the PostgreSQL/Supabase database and backend data models for Jamory, an in-house app connecting students with music tutors, using the Supabase CLI to manage schema migrations and configure user auth/login flows.

Software Engineering Intern — Clustr Jan 2026 – Sept 2026 (expected)

Cross-platform React Native social/scheduling app | ~8-person dev team

- Built server-side community infrastructure using Firebase Cloud Functions, including CRUD operations for communities, posts, and comments with enforced ban/edit permissions to prevent client-side database tampering, plus join/leave workflows and activity feed endpoints.

CAMPUS INVOLVEMENT

Teaching Assistant – Data Structures & Algorithms, Dept. of Computer Science, Stony Brook University Aug 2025 – Dec 2025

- Supported a 240-student Data Structures & Algorithms course through weekly office hours and recitations, guiding students through asymptotic analysis, recursion, and core data structures including graphs, trees, heaps, and hash tables.

PROJECTS

Diary Handwriting Analysis Python, PyTorch, Hugging Face Transformers, OpenCV

- Initially fine-tuned Microsoft's TrOCR handwriting recognition model with a custom OpenCV line-segmentation preprocessing pipeline (grayscale conversion, binary thresholding, horizontal ink-projection profiling), reaching ~87% reliable page transcription on personal diary pages.
- Pivoted to DANIEL (Document Attention Network for Information Extraction and Labeling), a segmentation-free architecture that transcribes full pages directly and jointly extracts named entities, to surface topic and mood trends across diary entries from large volumes of handwritten data over time; currently in progress.

SKILLS

Languages: Python, C, Java, JavaScript, TypeScript

Databases & Cloud: PostgreSQL, MongoDB, Firebase/Firestore, AWS, GCP, Supabase

Frontend: React, React Native, Next.js

Backend: Node.js, Express.js, FastAPI

Tools: Git, Docker, Linux, PM2, Selenium